Langqi Xing

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PROFESSIONAL SUMMARY

Ph.D. researcher in microfluidics, colloidal and non-colloidal suspensions, granular flows, impact dynamics, soft matter, and transport phenomena with 5+ years of cross-disciplinary experience spanning experiments, multiphysics modeling, and scientific computing. Expertise includes microfluidic device design and fabrication, photolithography and soft lithography, fluorescence imaging, colloidal transport, electrokinetic interpretation, and quantitative analysis of complex fluid and particle-transport systems. Doctoral research combines device fabrication, optical characterization, particle tracking, scaling analysis, and physics-based modeling to understand transport behavior in confined and multiphase systems.

Experienced in connecting mechanical and chemical engineering principles with computational tools, including COMSOL-based multiphysics and thermal simulations, MATLAB/Python image processing, particle tracking, quantitative data analysis, and visualization. Hands-on experience with DLS, zeta-potential analysis, fluorescence-based measurements, and colloidal transport studies provides a strong foundation for linking microfluidics, soft matter, colloids, and simulations with real physical systems. Targeting highly technical R&D roles in complex fluids, AI-driven multiphysics modeling, lab-on-chip systems, biomaterials, jetting, coating, additive manufacturing, and related multiphase/multiphysics platforms.

EDUCATION

Northeastern University — *Boston, MA*
Ph.D., Mechanical Engineering (Expected Graduation: Dec 2027) — Jan 2023 – Present
M.S., Mechanical Engineering — Apr 2025

- Selected graduate coursework: Multiphase Transport, Multiscale Flow, Transport Phenomena, Nonequilibrium Physics, Chemical Engineering for Biosystems and Biomaterials, Convective Heat Transfer, Heat Conduction and Radiation

University of California, Santa Barbara — Sep 2018 – Dec 2022
B.S., Chemical Engineering — *Santa Barbara, CA*

- Selected coursework: Separation Processes, Soft Materials, Thermodynamics, Fluid Mechanics

SKILLS

Programming & Scientific Computing: MATLAB, Python, ImageJ, Power BI, LabVIEW
Experimental Instrumentation: fluorescence microscopy, zeta-potential measurements, photolithography and soft lithography, microfluidic device fabrication, 3D printing, microparticle, droplet, and soft-material fabrication, high-speed imaging, X-ray imaging, rheometry, laser cutting, Arduino, waterjet cutting
Modeling & CFD Simulation: COMSOL Multiphysics, Aspen HYSYS, SolidWorks, Adobe Illustrator
Languages: English, Mandarin Chinese

RESEARCH PROJECTS

Ph.D. Researcher — Electrokinetic Transport in Microfluidic Systems — Jan 2023 – Present
Tang Lab, Northeastern University — *Boston, MA*

- Designed and fabricated dead-end microfluidic devices using PDMS soft lithography, 3D-printed molds, and boundary-modification strategies to interrogate colloidal transport under electrolyte gradients.
- Executed fluorescence-microscopy-based diffusiophoresis and diffusio-osmosis experiments in confined geometries, quantifying boundary-condition-dependent transport, particle focusing, and delivery behavior.
- Developed MATLAB and ImageJ workflows for background subtraction, particle tracking, trajectory extraction, and transport-metric quantification from long-duration time-lapse datasets.
- Built analytical and physics-based models for solute diffusion and electrokinetic transport to interpret regime behavior and generate publication-quality maps, trends, and design guidance.
- Expanded the platform to hydrogel-based solute-gradient generators for sustained transport and programmable colloidal banding, integrating experiments, modeling, and technical communication across multiple projects.

Ph.D. Researcher — High-Speed X-Ray Imaging of Impact-Induced Jamming — Jan 2026 – Present
Argonne National Laboratory / Northeastern University — *Lemont, IL / Boston, MA*

- Prepared high-speed X-ray imaging experiments at Argonne National Laboratory to investigate suspension-droplet microstructure during impact-induced jamming, relaxation, and phase-transition-like behavior.

- Designed the study to reveal internal particle-scale dynamics during the earliest impact stages, linking transient droplet solidification and relaxation to liquid-pool deformation and cavity response.

Undergraduate Researcher — Complex Fluids and Suspension Transport Dec 2020 – Dec 2022
Sauret Lab, University of California, Santa Barbara *Santa Barbara, CA*

- Investigated capillary-driven coating, fiber alignment, and particulate suspension transport in confined geometries through microscopy-based experiments and quantitative image analysis.
- Used MATLAB and ImageJ to extract flow profiles, coating dynamics, and microstructural trends from experimental datasets, supporting mechanistic interpretation of fluid and particulate behavior.
- Contributed to studies on particle adhesion, confined-suspension rheology, and granular collapse, supporting multiple peer-reviewed publications in fluid mechanics and soft-matter journals.

Undergraduate Research Assistant — Sustainability Modeling Nov 2021 – Dec 2022
C-Thru Global Project, University of California, Santa Barbara *Santa Barbara, CA*

- Quantified energy consumption and emissions across global chemical supply chains to evaluate process-level sustainability and decarbonization opportunities.
- Conducted literature reviews and developed mass and energy balance models for chemical production pathways, translating process data into comparative environmental metrics.
- Supported cross-disciplinary research connecting chemical engineering analysis with industrial systems, supply chains, and broader sustainability objectives.

Undergraduate Researcher — Thermal Reactor Modeling Dec 2020 – Jun 2021
McFarland Lab, University of California, Santa Barbara *Santa Barbara, CA*

- Performed heat-conduction and convection calculations for filament-reactor configurations to assess transport limitations and operating windows.
- Built COMSOL Multiphysics simulations of reactor temperature fields and used model results to analyze thermal gradients, energy efficiency, and design sensitivity.

PUBLICATIONS

- **L. Xing**, X. Tang. 'Controlling particle dynamics in dead-end channels via boundary effects." *Nanoscale*, Apr 2026.
- R.-S. Sharma, W. Sarlin, **L. Xing**, C. Morize, P. Gondret, A. Sauret. 'Effects of interparticle cohesion on the collapse of granular columns." *Physical Review Fluids*, Jul 2024.
- D.-H. Jeong, **L. Xing**, M. Ka Ho Lee, N. Vani, A. Sauret. 'Deposition and alignment of fiber suspensions by dip coating." *Journal of Colloid and Interface Science*, Jun 2023.
- D.-H. Jeong, **L. Xing**, J.-B. Boutin, A. Sauret. 'Particulate suspension coating of capillary tubes." *Soft Matter*, Oct 2022.

CONFERENCES

Engineering Tunable Hydrogel Beacon Configurations for Extended-Range Colloidal Transport Nov 2025
2025 AIChE Annual Meeting *Boston, MA*

- Presented a microfluidic strategy to generate sustained electrolyte gradients for long-range particle delivery into dead-end channels.
- Quantified boundary-material-dependent diffusio-osmotic backflow and separatrix shifts to map transport regimes and inform device design.

Controlling Particle Dynamics in Dead-End Channels via Tunable Boundary Effects Nov 2024
77th Annual Meeting of the Division of Fluid Dynamics *Salt Lake City, UT*

- Explored diffusiophoresis-driven colloidal transport under electrolyte gradients, focusing on boundary-driven fluid flows.
- Demonstrated manipulation of particle trajectories by modifying wall streaming potentials at dead-end boundaries.

Leveraging Boundary Effects for Particle Delivery into a Dead-End Channel Jun 2024
99th New England Complex Fluids Workshop, Brown University *Providence, RI*

- Presented a microfluidic strategy for leveraging boundary-material effects and electrolyte gradients to guide colloidal particle delivery into dead-end channels.
- Explained how tunable streaming potentials and boundary-driven flows influence particle trajectories, separatrix formation, and controlled transport behavior in confined geometries.

POSTERS

Engineering Tunable Hydrogel Beacon Configurations for Extended-Range Colloidal Transport Dec 2025
Northeastern University MIE Graduate Research & Art Expo *Boston, MA*

- Presented a research poster and associated visualization materials on "chemical beacon" designs that generate sustained gradients for pump-free particle delivery.
- Communicated key mechanisms and performance trends using publication-quality graphics and flow visualizations.

Controlling Particle Dynamics in Dead-End Channels via Tunable Boundary Effects Feb 2025
AAAS Annual Meeting *Boston, MA*

- Delivered an interactive e-poster to scientific audiences on boundary-tuned particle delivery in dead-end microchannels.
- Discussed how different boundary materials shift particle turning behaviors, supported by experiments and theory.

WORK EXPERIENCE

R&D Digital Transformation Intern — Ingredient Data Infrastructure Jul 2021 – Sep 2021
PepsiCo *Shanghai, China*

- Led the collection, cleaning, and standardization of approximately 10,000 ingredient specifications to improve traceability, searchability, and data integrity across R&D workflows.
- Developed interactive dashboards and structured data views to accelerate ingredient screening, technical review, and formulation decision-making.
- Collaborated with cross-functional teams to establish a centralized ingredient library supporting innovation projects and digital knowledge management.

Marketing Data Analyst Intern — Polymer Product Analytics Jul 2021
ExxonMobil *Shanghai, China*

- Analyzed polymer-product performance and market data to identify commercially relevant opportunities, customer-facing insights, and potential cost-saving strategies.
- Built and maintained an internal service platform to streamline communication among suppliers, technical teams, and business stakeholders.
- Presented data-driven analysis to senior management to support strategic marketing and product-positioning decisions.

Physician Assistant Intern — Orthopedic Surgery and Biomarker Studies Jun 2018 – Aug 2020
Drum Tower Hospital *Nanjing, China*

- Supported cruciate-ligament reconstruction studies using animal models by assisting with surgical preparation, sterile technique, and experimental coordination.
- Performed DNA sequencing and biomarker-related laboratory analyses to investigate bone-healing mechanisms and monitor post-surgical outcomes.

Research Associate Intern — Pharmaceutical Characterization Sep 2019 – Jan 2020
Jiangsu Kangyuan Pharmaceutical Co. *Jiangsu, China*

- Operated HPLC systems to quantify the purity of "Compound Nanxing Pain Paste," a pharmaceutical formulation used for osteoarthritis pain treatment.
- Analyzed physical and chemical properties using mass-spectrometry data to support formulation characterization and quality evaluation.

LEADERSHIP & AWARDS

Best Presentation Award Recipient Nov 2024
Research Expo, Northeastern University *Boston, MA*

Vice President & Treasurer Sep 2018 – Sep 2021
AIChE UCSB Chapter *Santa Barbara, CA*

- Planned and hosted professional-development activities and technical workshops with industry speakers and faculty.
- Managed chapter finances, monitored expenditures, and helped secure sponsorships to support student outreach.
- Coordinated operations that strengthened engagement within the chemical engineering student community.

World Champion Nov 2015
FRC Robotic Challenge, World Adolescent Robot Contest *Beijing, China*